

KARAN RAJENDRA

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SUMMARY

Software Engineer with 3+ years of experience designing and deploying **scalable backend architectures** for **E-commerce, FinTech, and Enterprise IT domains**. **Architected and optimized distributed systems** using **Java, Spring Boot, Kubernetes, Docker, and AWS**, achieving **sub-100ms API response times for 100,000+ concurrent users**. Experienced in **event-driven microservices, CI/CD pipelines, and fault-tolerant data storage**, driving measurable performance gains and reliability for critical business applications.

EXPERIENCE

Research Assistant – Distributed Systems & ML Infrastructure

Stony Brook University | USA

June 2025 – Present

- **Engineered** a distributed **ML** data pipeline using **Python** and **Kubernetes**-based scheduling, accelerating model training throughput by **500** samples per second for the core research team.
- Architected fault-tolerant microservices with **Kafka** for **event streaming** and **gRPC** for **communication**, containerized in **Docker** to maintain **data integrity** during weekly chaos engineering tests.
- Optimized a large-scale distributed training job by profiling with **perf** and **Valgrind**, reducing its **completion time** from **5 hours to 4 hours** without altering the **model's accuracy**.
- Automated a multi-node CI/CD pipeline with **GitHub Actions** and **Prometheus monitoring**, slashing **environment setup time** for **new team members** from **2 hours to 30 minutes**.

Graduate Teaching Assistant

Stony Brook University | USA

August 2024 – December 2024

- Instructed a cohort of **60+ graduate students** on core distributed systems principles, including **consensus algorithms (Raft)**, **data sharding**, and replication strategies for fault-tolerant architectures.
- Architected and **debugged 5+** hands-on programming assignments using **gRPC**, requiring students to implement and analyze trade-offs in **consistency, availability, and partition tolerance**.
- Mentored students through **complex system design challenges**, elevating the cohort's **average score from B- to B+** across **5 programming assignments** and fostering **collaborative skills**.

Software Engineer

Dell Technologies | INDIA

January 2021 – June 2023

- **Spearheaded** the design and implementation of **Java-based microservices architecture** using **Spring Boot**, enhancing system reliability and reducing bottlenecks for Dell's multi-region enterprise infrastructure platforms.
- **Revamped React.js** front-end interfaces for cloud dashboards, improving user navigation and accessibility for thousands of internal and client users, while ensuring responsive, cross-browser compatibility.
- **Orchestrated** integration of **RESTful APIs** with **Swagger-documented endpoints**, enabling seamless service-to-service communication and real-time insights for infrastructure monitoring and telemetry systems.
- **Modernized PostgreSQL database queries** and schema structures, reducing average latency by over **50%** and enabling near real-time analytics for cloud resource utilization and reporting.
- **Automated** deployment of containerized applications using **Docker** and **Jenkins CI/CD pipelines**, ensuring zero-downtime releases, consistent build environments, and faster delivery cycles for internal software tools.
- **Collaborated** with cross-functional teams to develop **cloud-native applications on AWS (EC2, S3, Lambda)**, improving uptime and supporting proactive monitoring for Dell's device telemetry systems.

TECHNICAL SKILLS

Languages & Programming: Java , JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Frameworks & Libraries: Spring Boot, Hibernate, JPA, React.js, Angular, Bootstrap, SASS, jQuery

Backend & APIs: RESTful APIs, Spring MVC, Node.js (secondary), Express.js, Swagger UI, Postman

Databases: PostgreSQL, MongoDB, Cassandra

Cloud & DevOps Tools: AWS (EC2, Lambda, S3), Azure DevOps, Google Cloud Run, Docker, Git, GitHub, GitHub Actions, Jenkins

Testing & Debugging: JUnit, Mockito, Postman, Unit Testing, Form Validation

CI/CD & Deployment: GitHub Actions, Jenkins, Docker, Azure Pipelines

Design & Prototyping: Figma, Responsive Web Design

Development Practices: Agile (Scrum), Git, Jira, Version Control

EDUCATION

Master of Science in Computer Engineering

Stony Brook University (SUNY) | NY

August 2023 – May 2025

B.Tech in Computer Science and Engineering

Visvesvaraya Technological University (VTU) | INDIA

August 2017 – August 2021

PROJECTS

Raft Consensus Engine: Devised a **production-grade Raft consensus engine** in **Go**, validating strong consistency and **sub-200ms leader** failover under network partitions via rigorous **Jepsen** testing to reliably synchronize distributed **ML workloads**.

Dynamo-Inspired Key-Value Store (C++): Built a highly-available distributed **key-value store** using consistent hashing and **vector clocks**, sustaining **50K ops/sec** at **10ms p99 latency despite 30%** node failures to serve as durable storage for critical **ML artifacts**.

Real-Time Analytics Pipeline (Kafka, Python, AWS): Configured a **fault-tolerant real-time analytics pipeline** ingesting **100K+ events/min** into **AWS**, implementing **backpressure handling** and **replication** to maintain less than **5 minutes** of downtime per month for low-latency business intelligence.